

README

E2E translation proof — DeepSeek EN→Serbian (run e2e_deepseek_20260614T130829)

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Real end-to-end, real-system, anti-bluff proof (§11.4.83 / §11.4.107) that the HelixTranslate pipeline translates a real document with a real LLM provider after this session's 70+ bug fixes — captured during a same-session real run.

Command (real system, real network)

```
go build -o build/unified-translator ./cmd/unified-translator
# DEEPSEEK_API_KEY sourced from ~/api_keys.sh into the env (never
printed/committed - §11.4.10)
./build/unified-translator \
  -i test/assets/crow_and_pitcher_en.txt \
  -o crow_sr.txt \
  -source-lang en -target-lang sr \
  -provider deepseek -model deepseek-chat -timeout 180s
```

Result: Translation completed successfully | provider=deepseek
model=deepseek-chat duration=2.89s (exit 0).

What was produced

- A **valid EPUB** (the CLI emits EPUB): mimetype (application/epub+zip) + META-INF/container.xml + OEBPS/content.opf + OEBPS/toc.ncx + OEBPS/chapter1.xhtml. Raw artifact: qa-results/e2e_deepseek_20260614T130829/crow_sr.txt (git-ignored).
- Source (EN): source_en.txt
- Translated chapter text extracted from the EPUB (SR Cyrillic): translated_chapter_sr.txt
- Provider session report: session_report.md

Anti-bluff verification (§11.4.107)

Check	Result
Real provider call	DeepSeek deepseek-chat, 2.89s, exit 0
Output is real Serbian Cyrillic	205 Cyrillic characters ("Гавран и крчаг ... Жедан гавран наиђе на крчаг...")
Differs from source	yes — no English source phrase ("thirsty crow") present
No placeholder/bluff text	yes — no TODO/placeholder/"translation failed"/"xlate("
Valid output ebook	yes — well-formed EPUB zip with mimetype + OPF + NCX + chapter
No credential leak	yes — grep of the API key over all evidence files = clean (§11.4.10)

Source: A thirsty crow found a pitcher with a little water... →

Translated: Жедан гавран наиђе на крчаг с мало воде на дну...

Provider coverage note (operator action)

`~/api_keys.sh` provides DeepSeek, Gemini, Zhipu, Groq, Mistral, OpenRouter, +~30 more — but **NOT OpenAI or Anthropic**. To prove/enable those providers end-to-end, add `OPENAI_API_KEY` / `ANTHROPIC_API_KEY` to `~/api_keys.sh`.

This run also surfaced and fixed a real CLI bug: `cmd/unified-translator` only accepted the `-api-key` flag at its validation gate and ignored the provider `*_API_KEY` env vars (which `resolveProviderAPIKey` already supported and the docs advertise) — now the gate falls back to the env var.